

**The Meaningful Information Limits of Generalized AI Structures; Inferred
Through the Epistemic Limitations, Constraint Satisfaction, and Information
Theory.**

Huxley Lewis

Think Global School

Capstone

Sophie Yang

November 2025

The Meaningful Information Limits of Generalized AI Structures; Inferred Through the Epistemic Limitations, Constraint Satisfaction, and Information Theory.

Artificial Intelligence, is an important part of the modern world, and as architectures are changing, and models are becoming increasingly larger, more general, and intelligent, it becomes important to have an intuitive model for the kinds of problems that Artificial Intelligence can solve. This paper aims to provide such a framework for dismissing problems as impossible to solve with Artificial Intelligence systems that is both workable and intuitive, incorporating a philosophic perspective, as well as a mathematical rigour.

Artificial Intelligence, or AI, has grown significantly over the last decade with the rise of large language models and other foundation models (Bommasani et al., 2021). These models are examples of neural networks, which learn from data rather than being programmed with exact algorithmic behaviours (LeCun et al., 2015). Scaling studies of language models suggest that performance improves with additional parameters, data, and compute, while also making the cost of training and inference a central practical constraint (Hoffmann et al., 2022; Kaplan et al., 2020). Although the exact internal mechanisms by which LLMs generalize across new tasks remain only partially understood (Bommasani et al., 2021). A model for the structure of AI problems, and by extension their limitations, then, can provide insight into what we can achieve with these structures.

Epistemology is the study of human knowledge (Steup et al., 2024). It has also been used in recent work to analyse AI systems as black boxes and to reason about the limits of explanation, justification, and understanding in machine learning (Carabantes, 2020; Funer, 2022). Because data is a fundamental element of AI systems, epistemic questions about what is known, how it is learned, and what remains inaccessible apply directly to AI. The different methods by which an AI system acquires and represents data also shape its structure, including the contrast between symbolic and connectionist approaches (Russell et al., 1995).

Constraint Satisfiability Problems, or CSPs, are a general framework for representing

problems in terms of variables, domains, and constraints (Mackworth, 1977). This abstraction makes CSPs useful for discussing classes of problems independently of the details of any particular solver (Dechter, 1988). CSPs have long been used in symbolic AI to represent relational structure in domain-specific problems while supporting general solving procedures (Dechter, 1988; Mackworth, 1977). This style of representation supports mathematical arguments that may extend to AI systems when those systems can be modelled in CSP-like terms.

Information theory is a framework for quantifying information, communication, and uncertainty (Shannon, 1948). In machine learning, information-theoretic methods have been used to study how representations preserve, compress, and transform information during learning (Shwartz-Ziv & Tishby, 2017; Tishby & Zaslavsky, 2015). Information theory therefore offers one way to analyse the internal behaviour of contemporary connectionist models (Shwartz-Ziv & Tishby, 2017).

These three perspectives, epistemology, constraint satisfiability, and information theory, can be combined to provide a framework for understanding the meaningful information limits of generalized AI models. By analysing the constraints that govern the processing of input and generation in AI systems, and by quantifying the mutual information between input and output, we can determine whether meaningful information can be generated by an AI system in a given context. This framework can be applied to various thought experiments and real-world problems to assess the capabilities and limitations of AI systems in generating meaningful information. It presents a structure of AI that is not based on the specific architecture or implementation of the AI system, but rather on the fundamental principles that govern the relationship between input, processing, and generation. This allows for a more general analysis of AI systems and their limitations, regardless of the specific technologies or algorithms used in their construction. This structural view of artificial intelligence motivates the use of constraint satisfiability problems, information theory, and epistemology when reasoning about the meaningful information

limitations of generalized AI models.

Literature Review

Defining AI

Turing, in the imitation game, describes AI as being any algorithm that aims to replicate human’s mental faculties, instead of their physical qualities (Turing, 2016). Other definitions aim to avoid redundancy by specifying how algorithms process data as prediction machines (Russell et al., 1995). This provides a much better basis for understanding what AI is, some introductory courses to the field will describe AI as general function approximation algorithms (LeCun et al., 2015). This has been expanded in recent years as AI models have become successful in problems that were not previously understood as functions (Bommasani et al., 2021), such as natural language processing. The ability to gain meaningful information moves beyond pure numeric function approximation has incurred the use of definitions of AGI, or artificial general intelligence, to differentiate simpler AI from more capable AI models. AGI can similarly be defined in many ways, but, as its name implies, is a generalization of AI, where all mental faculties of people are replicated, this was popularized as a concept by Alan Turing, as he describes the Turing test (Russell et al., 1995; Turing, 2016), whereas a human interacts with an algorithm the human cannot differentiate if an output is the result of an algorithm or another human.

These conflicting definitions are now being shaped by current use of AI as the public are maintaining relationships with the field. This motivates the analysis of these definitions for their applicability in supporting modern uses of AI. The aforementioned definition of AI as automation of mental faculties, in this context, gains meaning since it provides implied use cases for AI, as a tool to replace their mental faculties. The definition of AI as a prediction machine can be introduced to a member of the public to provide understanding of the mechanisms by which AI automates mental processes. The final definition of AI pertinent, here, is that of the general function approximator, where it again implies a certain use case, that it be used to regress or approximate functions, without the condition of

knowledge of the structure of the function.

These standard definitions excel at providing intuitive understanding for a lay person to grasp what AI is, and how it can be used, but they fall short in the ability to capture the limitations of AI. This motivates the position here, that the definition for AI should be structural, instead of useful, and to provide an intuitive construction of this definition.

Epistemic Limitations

Knowledge has been defined as a justified true belief (JTB), although this has been challenged by Gettier style cases (Steup et al., 2024), as it does not account for cases such as accidental knowledge. This process of knowledge acquisition has many perspectives. Two of these perspectives are that of the empiricists and rationalists. Philosophers such as Hume, argue that knowledge is gained through experience, giving us more evidence, thus increasing the likelihood or probability that what one believes is true, justifying it based on inference from the ability to sense reality (Markie & Folescu, 2023). Descartes, alternatively, in his meditations on first philosophy, claims that the only knowledge that one can have is that which is infallible, constructing his method of doubting to conclude that his ability to doubt is the only infallibility. These can be understood as different premises of the JTB theory, where Descartes argues that justification and truth can only come through rationality, Hume argues that justification and truth come from the experience of reality. These perspectives ask "what can one know?" in the context of their given reality. In AI problems, the reality must be provided, this is a consequence of the fact that the user of the AI structure, does, in asking their question define its scope, and therefore its reality.

Newcomb's problem (Nozick, 1969), where one is tasked with understanding if future events are predestined, is treated as an example of the unknowable. This kind of thought experiment, where the notions of knowledge are challenged, is important for understanding the limits of AI, since it provides a case to better understand the limits of what can be known, and therefore what can be generated by an AI system. The Chinese Room (Obermeier, 1983) is another thought experiment that challenges the understanding of

knowledge and understanding in the context of AI systems. It raises questions about whether an AI system can truly understand the information it processes, or if it is simply manipulating symbols without any real comprehension. These thought experiments highlight the importance of understanding the epistemic limitations of AI systems, and how they relate to the meaningful information that can be generated by those systems.

Knowledge, as defined here is not sufficient to describe the use case of AI models, since knowledge is specific to the agent that understands the knowledge. Let meaningful information, then, be defined as a valid collection or arrangement of signs that convey meaning to a decoder. Where validity is defined as a constraint of the input and output and meaning is assumed¹to be captured in the signs of the domain, and signs are symbols that incur meaning beyond their literal shape (Thornbury, 2011).

Meaning

There are many theories of meaning. In semiotics, meaning is the relationship between a sign, its object, and its observer (Thornbury, 2011). There have been many scholars that aimed to concern themselves with the "problem of meaning". Most famously Charles Peirce, in his pragmatics theory, aimed to unite language, not as a function of human description, but as a tool for understanding the world. This is different for instance to other philosophers attempting this question. For instance, Wittgenstein, in his later philosophy, argues that meaning is a function of the use of language in a given context (Speaks, 2010). This is different from Peirce's theory, where meaning is a function of the relationship between signs and their objects. These theories of meaning have implications for how we understand the meaningful information that can be generated by AI systems. If meaning is a function of the use of language in a given context, then the meaningful information that can be generated by an AI system is limited by the context in which it operates. If meaning is a function of the relationship between signs and their objects, then the meaningful information that can be generated by an AI system is limited by the ability to establish valid relationships between signs and their objects. Other philosophers disregard

the concept of meaning, for instance, Heidegger, in his *Being and Time* argues that the question of meaning is a consequence of the question of being, and therefore, meaning is not a fundamental concept in understanding the world (Macquarrie, Robinson, et al., 1973). This paper takes the analytical perspective of Peirce, where meaning is a function of the relationship between signs and their objects, since it provides a more direct way to understand the meaningful information that can be generated by AI systems, and how it relates to the constraints of the input and output.

This concept has been rediscovered in the modern study of AI systems, under the field of informativeness, where meaning is ascribed in weighting of orderings of information, quantified by the observer. Informativeness, currently is being used as a tool to analyse the internal behaviour of contemporary connectionist models (Brockmeier et al., 2017), and to understand the limits of what can be generated by those models. This is important for understanding the meaningful information that can be generated by AI systems, since it provides a way to quantify the relationship between the input and output of an AI system, and how that relationship affects the meaningful information that can be generated.

Definition 0.1 (Meaningful Information). Meaningful information is a valid assignment of signs.

Defining Terms

Definition 0.2 (Constraint Satisfiability Problem). A constraint satisfaction problem (CSP) is a tuple $\mathcal{P} = \langle X, D, C, A \rangle$, where:

$$X := \{X_1, \dots, X_n\} \text{ is a set of variables } X_i. \quad (1)$$

$$D := \{D_1, \dots, D_n\} \text{ assigns a domain } D_i \text{ to each variable } X_i. \quad (2)$$

$$C := \{C_1, \dots, C_m\} \text{ is a set of constraints } c_i. \quad (3)$$

¹ In extensions of this theory, this assumption must be taken into account, that the problem might be solvable on some general AI model.

Each constraint $C_j \in C$ is a pair $C_j = \langle S_j, R_j \rangle$, where $S_j \subseteq X$ is the scope and R_j is a set of valid tuples $R_j \subseteq \prod_{x_i \in S_j} D_i$.

$$A := \left\{ \mathbf{x} \in \prod_{\ell} D_{\ell} : \forall c, \mathbf{x}_{S_c} \in R_c \right\}, |A| > 0 \text{ is the set of valid assignments } \mathbf{x}_k. \quad (4)$$

Definition 0.3 (Probability). Let $P_i(a)$ denote the probability $X_i = a$ or $\Pr(X_i = a)$ and be defined as,

$$P_i(a) = \frac{|\{\mathbf{x} \in A : x_i = a\}|}{|A|},$$

assuming a uniform distribution.

Let $P_{ij}(a, b)$ denote $\Pr(X_i = a, X_j = b)$, and be defined as,

$$P_{ij}(a, b) = \frac{|\{\mathbf{x} \in A : x_i = a, x_j = b\}|}{|A|}.$$

Definition 0.4 (Surprisal). The surprisal of an outcome a of random variable X is,

$$I_i(a) := -\log(P_i(a)),$$

where $p(x)$ denotes the probability of a .

Definition 0.5 (Entropy). ² The entropy of random variable X is the weighted average of the surprisal of all outcomes a in the domain D_i of X_i , and is defined as,

$$H(X_i) := - \sum_{a \in D_i} P_i(a) \log(P_i(a)).$$

2

The model of AI

As there are many definitions of AI, discussed above, there could be many constructions of models of problems for AI models. If it is assumed that some given object class of AI problems is definable on the basis of predicating its behaviours. Inductively there can be constructed certain properties of this class, from these definitions of AI,

² By convention, assume $0 \log(0) = 0$

intersectionality can be found to construct the predicates of this class. In all the definitions of AI, there is a common structure of input, processing, and generation. This structure can be used to construct a model for AI that captures the essential elements of AI systems while allowing for a wide range of specific implementations and architectures. This structure of input, processing, and generation is also consistent with the way that many connectionist AI systems are designed and implemented in practice, where data is collected and preprocessed as input, algorithms are applied to process that data, and outputs are generated based on the processed data. Therefore, this model of AI provides a useful framework for reasoning about the capabilities and limitations of AI systems in terms of their ability to generate meaningful information from their inputs.

Contextual Implications

This model has certain immediate implications for the kinds of problems that AI structures can solve. To reference earlier discussion on epistemic limitations, if the problem is not given a context, then, it can be inferred that there is no ability to justify any generation of meaningful information from any given input, since truth and justification are a consequence of the context. This can, by the assumption of a valid arrangement of signs, be inferred to mean that the meaning and context can be captured in the valid arrangement of signs of the domains of both the input and the generation. Therefore, the input and output domains must be definable into a domain of signs, and there must exist a way of arranging those signs in a valid manner. This implies that the input must be meaningful information by definition 0.1.

Relational Implications

This model also implies that the relationship between the input and the generation is crucial for determining the meaningful information that can be generated by an AI system, distinguishing generation from output. Since the ability to process the information from input to generation must exist. If there is no relationship between the input and the generation, then it is impossible for the AI system to process the input. On the other hand,

if there is a relationship between the input and the generation, then the AI system can generate meaningful information, but the amount of information that can be generated is limited by the strength of that relationship. If the relationship is too weak, then the AI system is under-constrained, meaning that the entropy at the output is not less than the entropy at the input, and information is not gained. If the relationship is too strong, then the AI system may be over-constrained, meaning that it cannot generate any output at all. Therefore, the meaningful information that an AI system can generate is limited by the balance of constraints that relate the input to the output.

Applying the mathematical model of AI

The results inferred in the previous sections can be justified on the basis of the definitions outlined in section .

This construction can begin with the input as a variable x with a domain D_x , and the generation as a variable y with a domain D_y . The processing, in totality, can be represented as the satisfaction constraint A^2 that relates x and y , by 0.1.

The relationship between the input and the generation must also be subject to the mutual information between x and y . Define mutual information in terms of the entropy of X and Y as,

$$I(X_i; X_j) = H(X_i) + H(X_j) - H(X_i, X_j).$$

or alternatively, in terms of the probabilities of X and Y as,

$$\forall \mathcal{P}, I(X_i; X_j) = \sum_{a \in D_i} \sum_{b \in D_j} P_{ij}(a, b) \log \left(\frac{P_{ij}(a, b)}{P_i(a)P_j(b)} \right).$$

and for $\mathcal{P}^2 = \langle \{\mathcal{X}, \mathcal{Y}\}, \{D_{\mathcal{X}}, D_{\mathcal{Y}}\}, \{A\}, A \rangle$,

$$I(\mathcal{X}; \mathcal{Y}) = \sum_{(x, y) \in A} \frac{|A_{xy}|}{|A|} \log \left(\frac{|A| |A_{xy}|}{|A_x| |A_y|} \right),$$

where,

$$A_x := \{\alpha \in A : \mathcal{X} = x\} \implies |A_x| = \sum_{y \in D_{\mathcal{Y}}} |A_{xy}|,$$

$$A_y := \{\alpha \in A : \mathcal{Y} = y\} \implies |A_y| = \sum_{x \in D_{\mathcal{X}}} |A_{xy}|,$$

$$A_{xy} := \{\alpha \in A : \mathcal{X} = x, \mathcal{Y} = y\} \implies |A_{xy}| \in \{0, 1\}.$$

Notably $I(X_i; X_j) = I(X_j; X_i)$, and $I(X_i; X_j) \geq 0$ in all cases, and $I(X_i; X_j) = 0$ iff X_i and X_j are independent.

Definition 0.6 (Dependence Condition). An output is only generation iff, $I(\mathcal{X}; \mathcal{Y}) > 0$. If $I(\mathcal{X}; \mathcal{Y}) = 0$, inputs and outputs are independent.

Theorem 0.1. For every CSP $\mathcal{P}^n = \langle X^n, D^n, C^n, A^n \rangle$ with $|X^n| = n \geq 2$, there exists a CSP with exactly 2 variables such that there exists a bijection $\Phi : A^n \leftrightarrow A^2$.

Proof. Induct on n .

Base case $n = 2$: $\mathcal{P}^2 \implies |X^2| = 2$ ■.

Inductive hypothesis: true for $n = k$.

Inductive step $(k + 1)$:

Let $\mathcal{P}^{k+1} := \langle X^{k+1}, D^{k+1}, C^{k+1}, A^{k+1} \rangle$.

Construct $\mathcal{P}^k := \langle X^k, D^k, C^k, A^k \rangle$ as follows.

Choose distinct $u, v \in X^{k+1}$, $u \neq v$ and let w be a new variable.

$$\text{Set } X^k := (X^{k+1} \setminus \{u, v\}) \cup \{w\}, \quad D_w := D_u \times D_v.$$

$$D^k := \{D_w\} \cup \{D_t : t \in X^{k+1} \setminus \{u, v\}\}.$$

Define

$$A^k := \left\{ \beta : \exists \alpha \in A^{k+1}, \beta_w = (\alpha_u, \alpha_v), \beta_t = \alpha_t \forall t \in X^{k+1} \setminus \{u, v\} \right\}.$$

Let $C^k := \{\langle X^k, A^k \rangle\}$

Define $\Phi : A^{k+1} \rightarrow A^k$ by $\Phi(\alpha) = \beta$, where

$$\beta_i = \begin{cases} (\alpha_u, \alpha_v), & i = w, \\ \alpha_i, & i \in X^{k+1} \setminus \{u, v\}. \end{cases}$$

Define $\Psi : A^k \rightarrow A^{k+1}$ by $\Psi(\beta) = \alpha$, where, if $\beta_w = (a, b)$,

$$\alpha_i = \begin{cases} a, & i = u, \\ b, & i = v, \\ \beta_i, & i \in X^{k+1} \setminus \{u, v\}. \end{cases}$$

For all $\alpha \in A^{k+1}$, set $\beta := \Phi(\alpha)$. Then

$$\Psi(\beta)_u = \alpha_u, \quad \Psi(\beta)_v = \alpha_v, \quad \Psi(\beta)_t = \alpha_t \quad \forall t \in X^{k+1} \setminus \{u, v\}.$$

Therefore $\Psi(\beta) = \alpha$, and $\Psi(\Phi(\alpha)) = \alpha$.

Take any $\beta \in A^k$ and set $\alpha := \Psi(\beta)$. Then

$$\Phi(\alpha)_w = \beta_w, \quad \Phi(\alpha)_t = \beta_t \quad \forall t \in X^{k+1} \setminus \{u, v\}.$$

Therefore $\Phi(\alpha) = \beta$, and $\Phi(\Psi(\beta)) = \beta$.

Hence $(\forall \alpha \in A^{k+1}, \Psi(\Phi(\alpha)) = \alpha) \wedge (\forall \beta \in A^k, \Phi(\Psi(\beta)) = \beta) \Rightarrow \Phi$ is bijective.

Therefore $\Phi : A^{k+1} \leftrightarrow A^k$, and by the inductive hypothesis, $\Phi : A^n \leftrightarrow A^2$. ■

Corollary 0.2 (Over-constraint). *For any $\mathcal{P}^2$, if x and y are mutually exclusive, then there exists no generation for any of the input's domain.*

Proof. Since for a given $\mathcal{P}^2$, iff $I(\mathcal{X}, \mathcal{Y}) = 0$ then there exists no generation for any of the input's domain.

If mutual exclusivity is granted, then $|A_{\mathcal{X}\mathcal{Y}}| = 0$ for all $x \in D_x$ and $y \in D_y$ by definition. Therefore, $I(\mathcal{X}; \mathcal{Y}) = \sum_{(x,y) \in A} \frac{0}{|A|} \log\left(\frac{|A|0}{|A_x||A_y|}\right) = \sum 0 \log(0) = 0$, by convention, and there exists no generation for any of the input's domain. ■

Corollary 0.3 (Under-constraint). *For any $\mathcal{P}^2$, if x and y are mutually occurring, then there exists no generation for any of the input's domain.*

Proof. Since for a given $\mathcal{P}^2$, iff $I(\mathcal{X}, \mathcal{Y}) = 0$ then there exists no generation for any of the input's domain.

If mutual occurrence is granted, then $|A_{xy}| = |A_x| = |A_y| = |A|$ for all $x \in D_{\mathcal{X}}$ and $y \in D_{\mathcal{Y}}$ by definition. Therefore, $I(\mathcal{X}; \mathcal{Y}) = \sum_{(x,y) \in A} \frac{|A|}{|A|} \log\left(\frac{|A||A|}{|A||A|}\right) = \sum 1 \log(1) = 0$ and there exists no generation for any of the input's domain. ■

Corollary 0.4 (Mapping). *If for a given $\mathcal{P}^2$, there exists a generation for every input, and there exists an input for every generation. If there exists an input that doesn't map to a generation, then there exists a positive mutual information between the input and the generation.*

Proof. Since for a given $\mathcal{P}^2$, iff $I(\mathcal{X}, \mathcal{Y}) = 0$ then there exists no generation for any of the input's domain.

If it is granted that there exists a generation for every input, and there exists an input for every generation, then $|A_{xy}| = 1$ for all $(x, y) \in A$ by definition. Therefore,

$$I(\mathcal{X}; \mathcal{Y}) = \sum_{(x,y) \in A} \frac{1}{|A|} \log\left(\frac{|A|}{|A_x||A_y|}\right).$$

For $I(\mathcal{X}; \mathcal{Y}) = 0$, it would require $\forall (x, y) \in A$, $|A| = |A_x||A_y|$ or $A = D_{\mathcal{X}} \times D_{\mathcal{Y}}$. However, since $\exists (x, y) \in A$, $|A_x| < |D_{\mathcal{Y}}|$ or $|A_y| < |D_{\mathcal{X}}|$, this condition cannot be satisfied. Therefore, $I(\mathcal{X}; \mathcal{Y}) > 0$ and there exists a positive mutual information between the input and the generation. ■

Tests and Analysis

For each of the following cases, the mathematical model of AI will be applied to determine the meaningful information limits of the problem. First, constructing the input, processing, and generation for each case, and then deriving the mutual information between the input and generation to determine whether meaningful information can be generated:

Case 0.1 (Newcomb's problem). Begin by defining a case of Newcomb's problem as a world where there are two boxes, one transparent and one opaque. The transparent box contains \$1000, and the opaque box contains either \$0 or \$1,000,000. A predictor, who is always correct, has predicted whether the player will take only the opaque box or both boxes. If the predictor predicts that the player will take only the opaque box, then it places \$1,000,000 in the opaque box; otherwise, it leaves it empty. The player then chooses whether to take only the opaque box or both boxes.

1. Construct the input $\mathcal{X}$ as the player's choice, with domain $D_{\mathcal{X}} = \{\text{take opaque box only, take both boxes}\}$.
2. Construct the generation $\mathcal{Y}$ as the predictor's prediction, with domain $D_{\mathcal{Y}} = \{\text{predict take opaque box only, predict take both boxes}\}$.
3. The processing can be represented as the set of constraints that relates the player's choice to the predictor's prediction, which are $C = \{\langle \{\mathcal{X}, \mathcal{Y}\}, A \rangle\}$, where A is the set of valid assignments that satisfy the constraints of the problem. In this case, the valid assignments are:

$$A = \{(\text{take opaque box only, predict take opaque box only}),$$

$$(\text{take both boxes, predict take both boxes})\}.$$

4. The mutual information between $\mathcal{X}$ and $\mathcal{Y}$ can be calculated as follows:

$$I(\mathcal{X}; \mathcal{Y}) = \sum_{(x,y) \in A} \frac{|A_{xy}|}{|A|} \log \left(\frac{|A| |A_{xy}|}{|A_x| |A_y|} \right).$$

In this case:

$$I(\mathcal{X}; \mathcal{Y}) = \frac{1}{2} \log \left(\frac{2 \cdot 1}{1 \cdot 1} \right) + \frac{1}{2} \log \left(\frac{2 \cdot 1}{1 \cdot 1} \right) = \log(2) > 0.$$

Therefore, there is a positive mutual information between the player's choice and the predictor's prediction, which means that meaningful information can be generated in this case.

Case 0.2 (Chinese Room). Begin by defining a case of the Chinese Room as a world where there is a person who does not understand Chinese, but is able to manipulate symbols according to a set of rules. The person receives input in the form of Chinese characters, processes them according to the rules, and generates output in the form of Chinese characters.

1. Construct the input $\mathcal{X}$ as the Chinese characters that the person receives, with domain $D_{\mathcal{X}}$ being the set of all possible Chinese character inputs.
2. Construct the generation $\mathcal{Y}$ as the Chinese characters that the person generates, with domain $D_{\mathcal{Y}}$ being the set of all possible Chinese character outputs.
3. The processing can be represented as the set of constraints that relates the input to the generation, which are $C = \{\langle \{\mathcal{X}, \mathcal{Y}\}, A \rangle\}$, where A is the set of valid assignments that satisfy the constraints of the problem. In this case, it can be assumed that for all outputs there exists a valid input, and for all inputs there exists a valid output, which means that $I(\mathcal{X}; \mathcal{Y}) > 0$ by 0.4. Therefore, there is a positive mutual information between the input and the generation, which means that meaningful information can be generated in this case.
4. This seems to go against the intuition of the Chinese Room thought experiment, which suggests that the person in the room does not understand Chinese and therefore cannot generate meaningful information. However, this is a consequence of the assumption that meaningful information is defined as a valid assignment of signs. It suggests that the person in the room does get meaningful information from the input, but does not understand it in the same way that a native Chinese speaker would. This highlights the importance of the definition of meaningful information and how it relates to the constraints that govern the processing of the input and generation.

Case 0.3 (Image Classification). Construct an image classification problem, where the input $\mathcal{X}$ is an image, and the generation $\mathcal{Y}$ is a label that classifies the image. The processing can be represented as the set of constraints that relates the input to the generation, which are

$C = \{\langle\{\mathcal{X}, \mathcal{Y}\}, A\rangle\}$, where A is the set of valid assignments that satisfy the constraints of the problem. In this case, assume that for all images there exists a valid label, and for all labels there exists a valid image, which means that, by 0.4, there is a positive mutual information between the input and the generation, which means that meaningful information can be generated in this case.

This is consistent with the intuition of image classification problems, where the input image contains meaningful information that can be processed to generate a label that classifies the image. However, the amount of meaningful information that can be generated is limited by the constraints that govern the processing of the input and generation, which is reflected in the mutual information between the input and generation. This highlights the importance of understanding the relationship between the input and generation in AI systems, and how it affects the meaningful information that can be generated.

Limitations of the model

This model provides a useful framework for reasoning about the capabilities and limitations of AI systems in terms of their ability to generate meaningful information from their inputs. However, there are some limitations to this model that should be acknowledged. The model assumes that the input and generation have multiple solutions, which may not be the case. In practice, it can be understood that single solution problems can be solved using AI models, and that capability is not captured fully. Additionally, the model lacks structure for understanding tractability, although, it can provide an estimate of tractability, using the definition that information gain is the definition of computation. Therefore, if the mutual information between the input and generation is positive, then there is some information gain, and therefore some computation that can be performed. However, this does not provide a precise measure of tractability, and further work would be needed to develop a more detailed understanding of the computational complexity of AI systems based on this model. This model also assumes informativeness, or the measure of the structure of the meaningful information. This is possible to be taken into account, through an informative

weighting of the mutual information, although it was outside the scope of this paper, and is left for future work. Additionally, this model does not account for the dynamic nature of AI systems, where the input, processing, and generation can change over time. This is an important aspect of many AI systems, and further work would be needed to develop a more dynamic model that captures these changes. Possibly through a conception of Markov chains in the structure of input to generation, with feedback.

This work is also limited greatly, by the specific definitions of AI, epistemology, and meaning that were chosen. Different definitions of these concepts could lead to different conclusions about the capabilities and limitations of AI systems. Therefore, it is important to consider the implications of different definitions and how they affect the analysis of AI systems based on this model. These definitions were chosen based on their relevance to the analysis of AI systems and their ability to capture the essential elements of AI, but further work could be done to explore alternative definitions and their implications for understanding AI systems. The bias towards analytical philosophy, also encourages further understanding into the implications of other philosophical perspectives, such as phenomenology or existentialism, on the analysis of AI systems and their capabilities and limitations, and how they might be related to other mathematical models of AI problems.

Conclusion

This paper has presented a model of AI as a structure of input, processing, and generation, and has applied this model to reason about the capabilities and limitations of AI systems in terms of their ability to generate meaningful information from their inputs. The model has been justified based on the definitions of AI, epistemology, and constraint satisfaction problems, and has been applied to several cases to demonstrate its usefulness in understanding the relationship between the input and generation in AI systems. While there are some limitations to this model, it provides a useful framework for reasoning about the capabilities and limitations of AI systems, and further work can be done to address these limitations and develop a more detailed understanding of the computational complexity of

AI systems based on this model.

This method of reasoning about AI systems in terms of their ability to generate meaningful information from their inputs can be applied to a wide range of AI problems, making use of more intuitive approaches to understand and formalize them and their constraints that govern the processing of the input and generation. By understanding these constraints, a clearer understanding of the capabilities and limitations of AI systems could be achieved, as well as further work towards developing more effective and efficient AI models that are able to generate meaningful information from their inputs. After adding structure for tractability and informativeness, this model can be used to provide a more detailed understanding of the computational complexity of AI systems, and can help to guide the ways in which AI algorithmic capabilities and limitations in the future are understood.

References

- Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., et al. (2021). On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*.
- Brockmeier, A. J., Mu, T., Ananiadou, S., & Goulermas, J. Y. (2017). Quantifying the informativeness of similarity measurements. *Journal of Machine Learning Research*, 18(76), 1–61.
- Carabantes, M. (2020). Black-box artificial intelligence: An epistemological and critical analysis. *AI & Society*, 35(4), 943–952.
- Dechter, R. (1988). Backtracking algorithms for constraint satisfaction problems: A survey. *Computational Intelligence*, 4(3), 273–289.
- Funer, F. (2022). Accuracy and interpretability: Struggling with the epistemic foundations of machine learning-generated medical information. *Philosophy & Technology*, 35(2), 1–28.
- Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., de Las Casas, D., Hendricks, L. A., Welbl, J., Clark, A., et al. (2022). Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*.
- Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020). Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.
- Mackworth, A. K. (1977). Consistency in networks of relations. *Artificial Intelligence*, 8(1), 99–118.
- Macquarrie, J., Robinson, E., et al. (1973). *Being and time*. Blackwell Oxford.
- Markie, P., & Folescu, M. (2023). Rationalism vs. Empiricism. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Spring 2023). Metaphysics Research Lab, Stanford University.

- Nozick, R. (1969). Newcomb's problem and two principles of choice. In *Essays in honor of carl g. hempel: A tribute on the occasion of his sixty-fifth birthday* (pp. 114–146). Springer.
- Obermeier, K. K. (1983). Wittgenstein on language and artificial intelligence: The chinese-room thought experiment revisited. *Synthese*, 339–349.
- Russell, S., Norvig, P., & Intelligence, A. (1995). A modern approach. *Artificial Intelligence. Prentice-Hall, Englewood Cliffs*, 25(27).
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423.
- Shwartz-Ziv, R., & Tishby, N. (2017). Opening the black box of deep neural networks via information. *arXiv preprint arXiv:1703.00810*.
- Speaks, J. (2010). Theories of meaning.
- Steup, M., Neta, R., & Turri, J. (2024). Epistemology. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Fall 2024). Metaphysics Research Lab, Stanford University.
- Thornbury, C. (2011). Peirce's general theory of signs. *Finding Meaning, Cultures Across Borders: International Dialogue between Philosophy and Psychology*, 49–57.
- Tishby, N., & Zaslavsky, N. (2015). Deep learning and the information bottleneck principle. *2015 IEEE Information Theory Workshop (ITW)*, 1–5.
- Turing, A. (2016). *The imitation game*. Channel 4.